

Main Topic

Fault-Tolerant Embedded Edge AI Systems

Subtopic

Multimodal Fault Detection and Reliable Activation in Assistive Sensing Systems

Tentative Title

Fault-Tolerant Multimodal Presence Verification and Safe Activation Policies on ESP32-S3 Using mmWave Radar and Vision

Abstract

Assistive room-scale embedded systems increasingly rely on edge perception to decide when to activate high-power sensing and interaction (camera inference, audio capture, actuation). However, these systems face frequent faults and uncertainties including lighting-induced vision degradation, radar clutter, transient communication errors, timing variability, and resource constraints. A single-modality activation scheme is therefore brittle and can produce two critical failure modes: (i) false activations that waste energy and may raise privacy concerns, and (ii) missed activations that deny assistance and reduce system trustworthiness. [1]

This project proposes a fault-tolerant activation subsystem implemented on an ESP32-S3 platform integrating a camera module and a 24 GHz FMCW mmWave radar module. The approach uses heterogeneous redundancy and system-level diagnosis: mmWave provides presence cues robust to lighting and can be privacy-preserving, while vision provides spatial confirmation and helps resolve radar ambiguities. The subsystem performs cross-sensor consistency checks and temporal redundancy to detect inconsistencies, classifies them into transient versus persistent faults, and selects recovery actions accordingly. [2]

The system incorporates fault detection mechanisms such as temporal redundancy, cross-sensor consistency checking, and anomaly detection in sensor outputs. Detected inconsistencies are further classified into transient and persistent faults. Based on this classification, the system employs adaptive fault-handling strategies. A two-tier safe-state mechanism is introduced: a soft safe state for temporary inconsistencies that triggers re-validation, and a hard safe state for persistent faults that blocks system activation, isolates unreliable inputs, and optionally notifies the user.

A two-tier safe-state policy is introduced. In a soft safe state, the system suppresses full activation and performs re-validation under a bounded time window; in a hard safe state (persistent inconsistency), the system blocks activation, isolates low-confidence inputs, and optionally provides user feedback while maintaining minimal power consumption. Design choices are guided by multi-source sensor fusion research that explicitly addresses cross-sensor inconsistency and uncertainty quantification, and by radar presence-detection research that motivates OOD-aware monitoring for cluttered indoor environments. [2]

Evaluation will combine controlled fault injection (sensor packet drops, radar framing errors, camera frame corruption/blur, induced timing delays) with environmental stress tests (low light, reflective surfaces, moving clutter). Metrics will include false activation rate, missed activation rate, time-to-detect inconsistency, fault coverage (fraction of injected faults detected), and activation latency/energy overhead relative to a non-fault-tolerant baseline.[3]

This work demonstrates how classical fault-tolerance principles (redundancy, error detection, diagnosis, and graceful degradation) can be concretely applied to practical edge AI activation pipelines on resource-constrained embedded hardware, improving reliability and trustworthiness for assistive room settings.

References used

- [1] [An Indoor People Counting and Tracking System using mmWave sensor and sub-sensors - ScienceDirect](#)
- [2] [A novel multi-source sensor correlation adaptive fusion framework with uncertainty quantification for intelligent fault diagnosis - ScienceDirect](#)
- [3] [Machine-learning assisted model-implemented fault injection](#)

Additional References

- [4] [\[2308.02396\] HOOD: Real-Time Human Presence and Out-of-Distribution Detection Using FMCW Radar](#)
- [5] [A Robust mmWave Radar Framework for Accurate People Counting and Motion Classification](#)

Other Resources

Espressif Systems. ESP32-S3 Series Datasheet (secure boot, cryptographic components, camera interface) and Technical Reference Manual.

ESP-IDF documentation for Secure Boot v2 and flash encryption on ESP32-S3.

Seeed Studio. 24 GHz mmWave Sensor for XIAO datasheet and user manual (FMCW principle, range gates, UART frame format).

GitHub repositories: esp32-camera and esp-tflite-micro for implementation feasibility.

Note on Scope

This CPRE 5450 project is intentionally scoped as a focused fault-tolerance study of the **activation/presence-verification subsystem**, taken from the broader CPRE 5750 multimodal assistive platform (which includes additional gesture/audio interaction and actuation features beyond what will be analysed in the fault-tolerant report).